

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fourth Semester of B. Tech. Theory Examination (CL)****April - May 2018****CL 215 Building and Town Planning****Date: 09.05.2018, Wednesday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. Separate Drawing sheets are provided for drawings.

SECTION – I**Q - 1 Do as directed:****[07]**

- i. Draw the symbol of tube light used in building drawing.
- ii. Write the full form for the abbreviation FSI used in building bye-laws.
- iii. The angle of light plane is ____ to _____.
 - a) 40° to 70° b) 45° to 63.5°
 - c) 30° to 60° d) 50° to 63°
- iv. No portion of room is assumed to be lighted if it is more than _____ m away from the opening.
 - a) 7.5 m b) 10 m
 - c) 12 m d) 5 m
- v. In _____ type of houses, a common boundary wall in form of structural barrier divides an independent plot into two units.
- vi. The temperature in the morning is less than the temperature at evening even though the rays of the sun are slanting. Justify.
- vii. Verandah should be provided in ____ and _____ direction to reduce exposure to the sun.

Q – 2.a Write a short note on ‘Aspect’ as a principle of building planning with neat sketch.**[04]****Q – 2.b** Write a short note on bye-laws regarding ‘Height of the Building Limitations’**[05]****OR**

Q – 2.b A plot owner proposed G + 1 construction with 175 m² built up area on each floor, on a plot of size 16 m x 21 m. If all side margins are 2 m, find the ground coverage and FSI proposed. If the FSI allowed in the area is 1.0, state with reasons whether the plan will be sanctioned or not. **[05]**

Q – 2.c Refer **Figure 1 (page 4)** and provide a plumbing layout for the same. Dimensions **[05]**

given in the drawing are in mm.

Q - 3 Answer any one.

[14]

- (i) The owner of a 12 m x 14 m rectangular residential plot wishes to construct a detached house on it. The longer side of the plot abuts a 9 m wide road and is having due East orientation. The built up area allowed is 60% of the plot area. The site is a level one. Design the single storey house with the following requirements.

- Living room
- Master Bedroom
- Children's Bedroom
- Kitchen with store
- Dining Room
- Common Toilet

Draw the plan to the scale of 1:100.

- (ii) Refer **Figure 1 (page 4)** and draw the section about I-J with a suitable scale assuming the necessary details. Provide all necessary levels in the drawing. Some of the specifications are as follows.

Level of plinth above the ground	= 450 mm
Thickness of wall in plinth	= 400 mm
Thickness of main walls	= 300 mm
Thickness of partition wall between bath and W.C	= 200 mm
Ceiling height for all rooms	= 3000 mm
Ceiling height for the verandah	= 2700 mm
Doors :	D = 1200 mm x 2100 mm
	D1 = 900 mm x 2000 mm
Windows :	W = 1500 mm x 1200 mm
	W1 = 1200 mm x 1200 mm
Ventilator :	V = 500 mm x 500 mm

Assume any other necessary data. All dimensions given in the drawing are in mm.

SECTION – II

Q - 4 Do as directed:

[07]

- i. An imaginary vertical transparent plane between the object and the observer on which the image of the object is projected and perspective is drawn is called _____.
 - a) Centre plane
 - b) Ground plane
 - c) Picture plane
 - d) Horizontal plane
- ii. In one point perspective the vertical lines remain _____ because they are

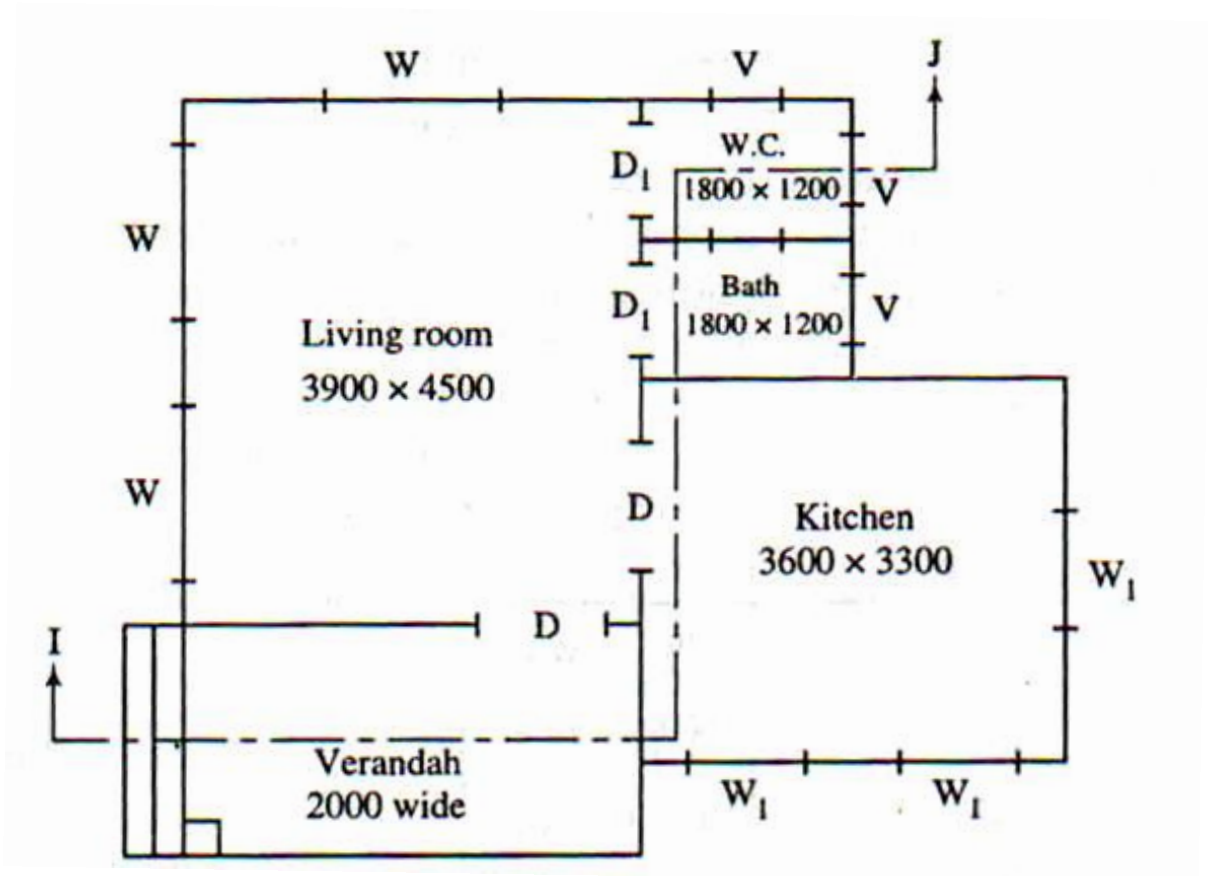


Figure 1 [Q-2. c & Q-3 (ii)]

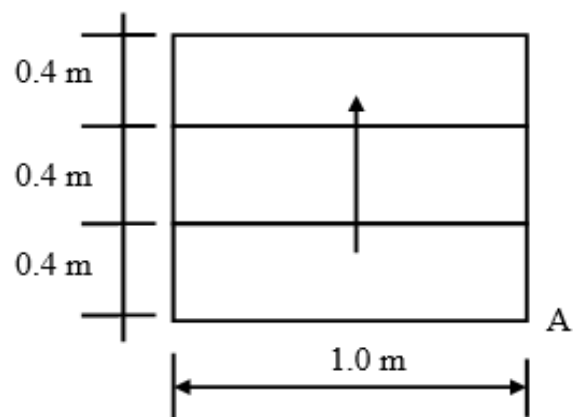


Figure 2 [Q-5. c (i)]

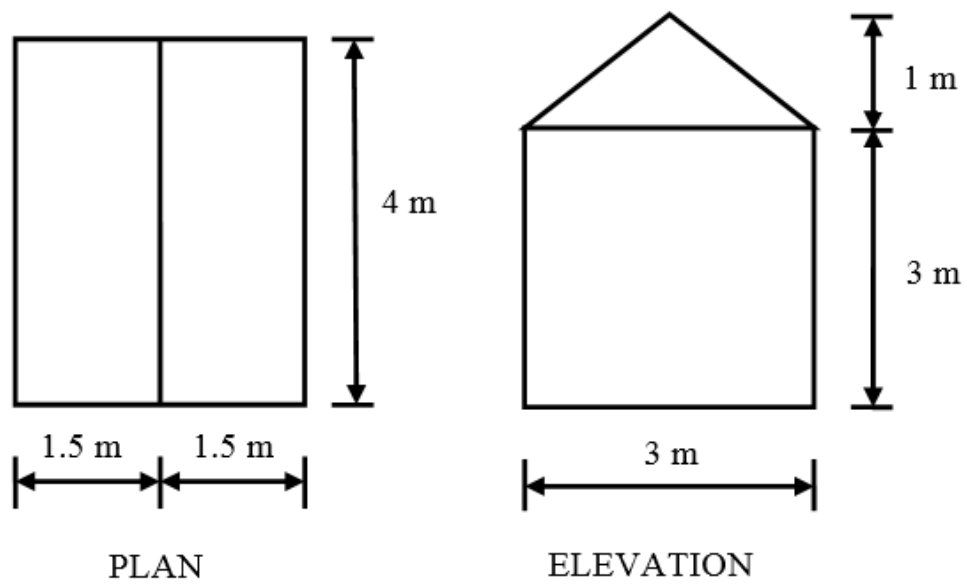


Figure 3 [Q-5. c (ii)]
